

This example demonstrates how traders might use a simple conditional to detect a potential bullish signal—
when price is above a moving average with higher-than-normal volume.

*?? **Did You Know?** The systematic hedge fund Renaissance Technologies, founded by mathematician James Simons, uses some of the most sophisticated conditional logic in the financial world. Their flagship Medallion Fund has averaged annual returns of approximately 66% (before fees) since 1988. A former employee once remarked that their trading systems contain thousands of nested conditional statements that evaluate market conditions across multiple timeframes simultaneously. These precisely defined conditions, coded as if statements and their variants, form the backbone of what many consider the most successful trading system in financial history.*

The `else` Statement: The Alternative Path

```
What if we want our program to do one thing if a condition is true, and something different if it's false? That's where the else statement 1. comes in.  
2. price = 150  
3. buy_threshold = 145  
4.  
5. if price < buy_threshold:  
6.     print("Buy signal generated!")  
7. else:  
8.     print("Price above threshold - no action required")
```

Since the price (150) is not below our threshold (145), Python executes the code in the `else` block, printing "Price above threshold - no action required."

Here's a more sophisticated trading example:

```
1. current_price = 42.75  
2. entry_price = 40.50  
3. stop_loss = 38.00  
4. target_price = 46.00  
5.
```